Group 114
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I. Pen-and-paper

1.)

$$\phi = \begin{bmatrix} 1 & 4 \\ 1 & 1 \\ 1 & 6 \\ 1 & 18 \\ 1 & 8 \end{bmatrix} \qquad z = \begin{bmatrix} 3.5 \\ 1.0 \\ 3.8 \\ 10.1 \\ 8.5 \end{bmatrix}$$

$$w_{OLS} = (\phi^T \phi)^{-1} \phi^T z =$$

$$= \left(\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 4 & 1 & 6 & 18 & 8 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 1 & 1 \\ 1 & 6 \\ 1 & 18 \\ 1 & 8 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 4 & 1 & 6 & 18 & 8 \end{bmatrix} \begin{bmatrix} 3.5 \\ 1.0 \\ 3.8 \\ 10.1 \\ 8.5 \end{bmatrix} = \begin{bmatrix} 1.46136 \\ 0.52955 \end{bmatrix}$$

$$\hat{z} = 1.46136 + 0.52955\, y_1 \times y_2$$

2.)

$$w_{Ridge} = (\phi^T \phi + \lambda I)^{-1} \phi^T z = \begin{bmatrix} 0.97319 \\ 0.56921 \end{bmatrix}$$

$$\hat{z} = 0.97319 + 0.56921\, y_1 \times y_2$$

$$\|w_{OLS}\|^2 = 2.41600 \quad , \quad \|w_{Ridge}\|^2 = 1.27110$$

Because of this regularization, which minimizes the sum of squared coefficients by adding a penalty term to the loss function, preventing the weights from becoming too large, we obtained smaller coefficients than the ones gotten when using OLS. This means the model learnt when using regularization is less likely to overfit.

3.)

For OLS:

$$MAE_{Train} = \frac{1}{5}\left(|3.5 - \hat{z}(2,2)| + |1.0 - \hat{z}(1,1)| + |3.8 - \hat{z}(3,2)| + |10.1 - \hat{z}(6,3)| + |8.5 - \hat{z}(8,1)|\right) =$$
$$= 1.12093$$

$$MAE_{Test} = \frac{1}{3}\left(|1.0 - \hat{z}(0,2)| + |6.2 - \hat{z}(3,4)| + |3.6 - \hat{z}(5,1)|\right) = 0.86214$$

For Ridge:

$$MAE_{Train} = 1.09458$$

$$MAE_{Test} = 0.61659$$

When using regularization, the test MAE is smaller than the train and also smaller than the test one when regularization wasn't used, showing that the model generalizes better when regularization is applied, which was expected. However, the train MAE when using regularization is lower than the train when using OLS, which was unexpected (meaning it should be higher), but explainable (for example, due to the small dataset).

4.)

Forward prop:

$$z^{[1]} = \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.2 \\ 0.2 & 0.1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.6 \\ 0.7 \end{bmatrix} \qquad x^{[1]} = \begin{bmatrix} 0.62246 \\ 0.64566 \\ 0.66819 \end{bmatrix}$$

$$z^{[2]} = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0.62246 \\ 0.64566 \\ 6.66819 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4.25016 \\ 3.58797 \\ 2.93637 \end{bmatrix} \qquad x^{[2]} = \begin{bmatrix} 0.56135 \\ 0.28777 \\ 0.15088 \end{bmatrix}$$

Derivatives:

$$\frac{\partial E(x^{[2]},t)}{\partial z^{[2]}} = x^{[2]} - t$$

$$\frac{\partial x^{[1]}(z^{[1]})}{\partial z^{[1]}} = \sigma(z^{[1]})(1-\sigma(z^{[1]})) = x^{[1]}(1-x^{[1]})$$

$$\frac{\partial z^{[i]}(W^{[i]}, b^{[i]}, x^{[i-1]})}{\partial W^{[i]}} = x^{[i-1]}$$

$$\frac{\partial z^{[i]}(W^{[i]}, b^{[i]}, x^{[i-1]})}{\partial b^{[i]}} = 1$$

$$\frac{\partial z^{[i]}(W^{[i]}, b^{[i]}, x^{[i-1]})}{\partial x^{[i-1]}} = W^{[i]}$$

Back prop:

$$\delta^{[2]} = \frac{E(x^{[2]},t)}{\partial z^{[2]}} = x^{[2]} - t = \begin{bmatrix} 0.56135 \\ 0.28777 \\ 0.15088 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.43865 \\ 0.28777 \\ 0.15088 \end{bmatrix}$$

$$\delta^{[1]} = \left(\frac{\partial z^{[2]}}{\partial x^{[1]}}\right)^T \cdot \delta^{[2]} \circ \frac{\partial x^{[1]}}{\partial z^{[1]}} = (W^{[2]})^T \cdot \delta^{[2]} \circ x^{[1]}(1-x^{[1]}) =$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} -0.43865 \\ 0.28777 \\ 0.15088 \end{bmatrix} \circ \begin{bmatrix} 0.62246 \\ 0.64566 \\ 0.66819 \end{bmatrix} \circ \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.62246 \\ 0.64566 \\ 0.66819 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ -0.03452 \\ -0.09725 \end{bmatrix}$$

$$\frac{\partial E}{\partial W^{[1]}} = \delta^{[1]} \left(\frac{\partial z^{[1]}}{\partial W^{[1]}}\right)^T = \delta^{[1]} (x^{[0]})^T = \begin{bmatrix} 0 \\ -0.03452 \\ -0.09725 \end{bmatrix} \begin{bmatrix} 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -0.06904 & -0.06904 \\ -0.19450 & -0.19450 \end{bmatrix}$$

$$W^{[1]} = W^{[1]} - \eta \frac{\partial E}{\partial W^{[1]}} = \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.2 \\ 0.2 & 0.1 \end{bmatrix} - 0.5 \begin{bmatrix} 0 & 0 \\ -0.06904 & -0.06904 \\ -0.19450 & -0.19450 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.1 \\ 0.13452 & 0.23452 \\ 0.29725 & 0.19725 \end{bmatrix}$$

$$b^{[1]} = b^{[1]} - \eta \frac{\partial E}{\partial b^{[1]}} = b^{[1]} - \eta\,\delta^{[1]} = \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix} - 0.5 \begin{bmatrix} 0 \\ -0.03452 \\ -0.09725 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.01726 \\ 0.14863 \end{bmatrix}$$

$$\frac{\partial E}{\partial W^{[2]}} = \delta^{[2]} \left(\frac{\partial z^{[2]}}{\partial W^{[2]}}\right)^T = \delta^{[2]} \left(x^{[1]}\right)^T = \begin{bmatrix} -0.43865 \\ 0.28777 \\ 0.15088 \end{bmatrix} \begin{bmatrix} 0.62246 & 0.64566 & 0.66819 \end{bmatrix} =$$

$$= \begin{bmatrix} -0.27304 & -0.28322 & -0.29310 \\ 0.17913 & 0.18580 & 0.19229 \\ 0.09392 & 0.09742 & 0.10082 \end{bmatrix}$$

$$W^{[2]} = W^{[2]} - \eta \frac{\partial E}{\partial W^{[2]}} = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} - 0.5 \begin{bmatrix} -0.27304 & -0.28322 & -0.29310 \\ 0.17913 & 0.18580 & 0.19229 \\ 0.09392 & 0.09742 & 0.10082 \end{bmatrix} =$$

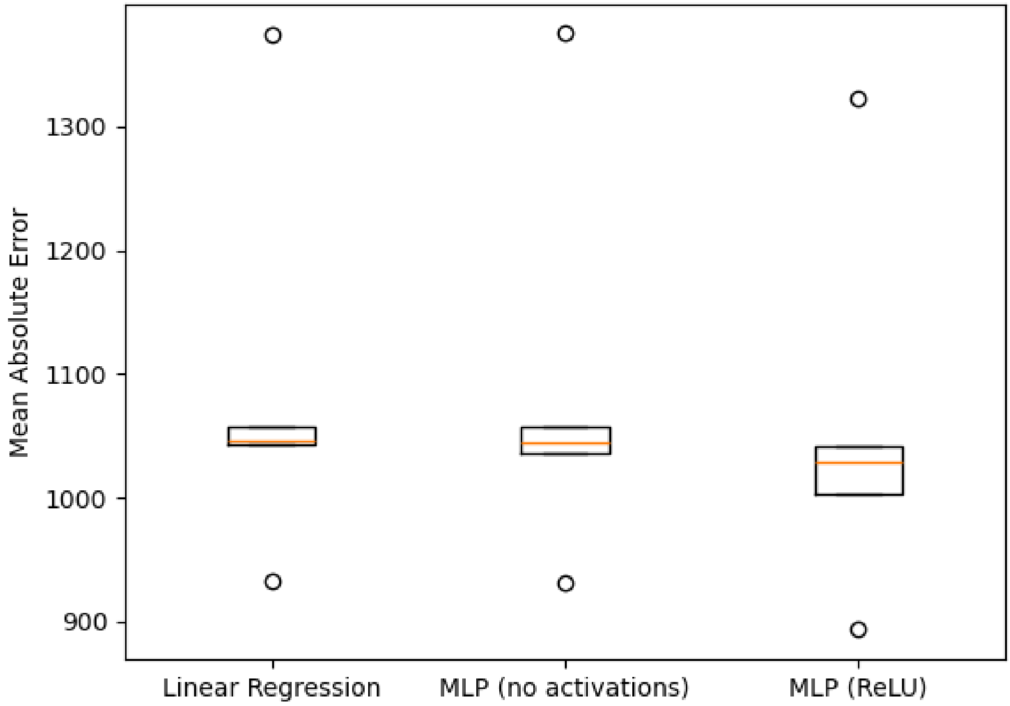
$$= \begin{bmatrix} 1.13652 & 2.14161 & 2.14655 \\ 0.910435 & 1.90710 & 0.90386 \\ 0.95304 & 0.95129 & 0.94959 \end{bmatrix}$$

$$b^{[2]} = b^{[2]} - \eta \frac{\partial E}{\partial b^{[2]}} = b^{[2]} - \eta \delta^{[2]} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - 0.5 \begin{bmatrix} -0.43865 \\ 0.28777 \\ 0.15088 \end{bmatrix} = \begin{bmatrix} 1.21933 \\ 0.85612 \\ 0.92456 \end{bmatrix}$$

When using a sigmoid activation, it allows the model to approximate any continuous function and to create non-linear decision boundaries (able to learn XOR and other non-linearly seperable problems). On the other hand, those things are not possible when no activation is used (can only create linear boundaries and its approximations are only limited to linear functions).

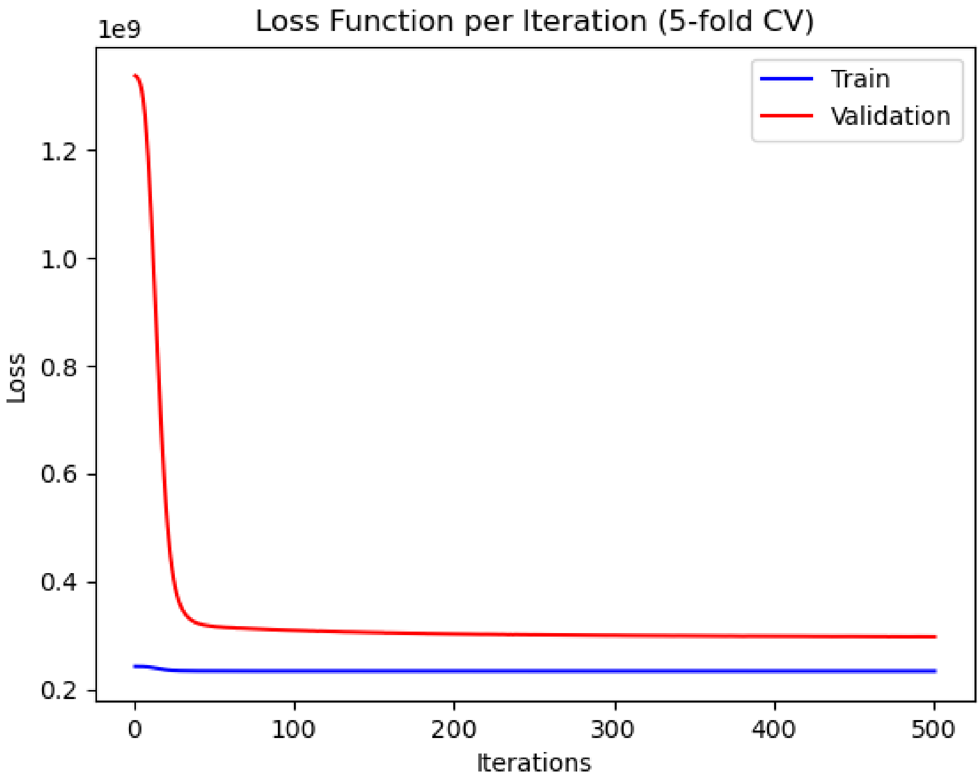
II. Programming

5.)



6.) A Linear Regression model can only capture linear relationships between the inputs and target which is almost identical to a MLP with no activations (hence the similarity of those models' boxplots), while a MLP with activation functions is able to learn non-linear surfaces and approximate any continuous function, enabling the model to learn complex relationships. In this case, the rent price seems to follow a non-linear trend with the features (supported by the lowest MAE mean and boxplot of the MLP with ReLU model).

7.)



We can see that the training loss starts with a low value and decreases slowly until it reaches a plateau. On the other hand, the validation loss starts much higher and decreases until hitting a plateau above the training loss. The gap between both losses at their respective plateau is significant, which means the model is overfitting to the training data: the model learns the training data very well but doesn't generalize to unseen data.